

(FireWire) port. The BIOS and the over-clocking options integrated into this board are a piece of art and few other motherboards have such customised BIOSes which are so easy to use.

What it all leads to...

This comparison test featured a gamut of chipsets and memory options—from Plain Jane motherboards to feature-packed performance power houses. There are lots of choices for every single type of mainstream processor and this test can be used to analyse how systems based on different processor/memory combinations would perform.

Pentium 4: If you want the maximum performance out of the Pentium 4, without giving much thought to the price, you have little choice but to opt for an RDRAM board. The **Gigabyte GA-8ITXE** came in trumps here; it even beat Intel's very own D850MD and is attractively priced as well. The DDR SDRAM boards performed admirably and the MSI 845 Ultra-ARU, ULTRA-C, ASUS P4B266 and Intel's D845BG are great DDR solutions. Nevertheless, they failed to dislodge the Gigabyte GA-8ITXE from the top spot. The DDR boards get severely hammered by their RDRAM counterparts in memory bandwidth-intensive applications like *Quake III Arena*, where they lose by a margin of about 20 per cent!

The Best Value Award goes to the **HIS P4M266 (M50-12)** board. This SDRAM board steadily chugged away in all our tests and offers the Pentium 4 solution at an unbelievably low price point. This board narrowly edged out the **Kranion KP1M845** (based on the Intel 845 SDRAM chipset) showing that the VIA P4M266 chipset is no pushover and performs pretty well when compared with the 845 SDRAM chipset.

Athlon XP/Duron: The battle for the performance crown in this category was cut-throat and the legendary **ASUS A7V266-E** came out victorious. It beat the MSI KT3 Ultra-ARU and the promising Soltek SL75DRV4 by a fair margin and is truly the board to have if you intend to push your Athlon XP to the max!

The SDRAM based boards are simply too slow for the faster Athlon XP processors and it's highly recommended to adopt a DDR solution on the lines of the VIA KT266A/KT333.

But if you are in the market for a real-

ly low-cost board for a Duron or a lower speed Athlon, then take a close look at the **Krypton M7VKS**. This board is based on the VIA Apollo Pro Savage KM133A chipset and costs only Rs 3,480. This price is tough to match and it's our Best Value winner.

Pentium III/Celeron (Tualatin): The Intel 815EP-based **ASUS TUSL2-M** has no peers here. It beat all the other boards hollow and is by far the most attractive platform for a new Tualatin processor.

The ultimate value proposition are the Intel 810E2 chipset-based motherboards with their integrated audio and video, and they allow for some really inexpensive systems. Once again an ASUS board takes the top honours here—the **TUWE-M** retails for only Rs 4,200 and its performance is not too shabby.

WINNERS!

Gigabyte GA-8ITXE

This is one of the best RDRAM-based boards you can buy. It performed brilliantly in *Quake III Arena*, churning out a whopping 361.8 fps. The GA-8ITXE fea-



Gigabyte GA-8ITXE: Brilliant combination of features and performance at a great price

tures a great number of overclocking options in the BIOS and even has onboard Ethernet support using the Realtek chip. It comes with a DualBIOS feature, which kicks in if the primary BIOS gets corrupted. The GA-8ITXE also includes EasyTune III, a nifty utility that allows for easy overclocking of the FSB within Windows itself.

The layout of this board is quite compact which makes installation a trifle difficult. The board ships with two C-RIMM terminators that can be used to terminate

Features that Count... and Others that don't

✓ **Overclocking made easy:** The Red-Storm Overclocking utility provided on Soltek boards is amazingly simple. It calculates the maximum extent to which a system can be overclocked without compromising on stability.

✓ **On-board Dolby Digital support:** An astounding feature found only in the nForce 420-D based motherboards (like the ASUS A7N266-E), it saves you the hassles of investing in an expensive add-on sound card to enjoy high-quality surround sound.

✓ **Integrated Ethernet:** One feature that almost every manufacturer should provide because it is relatively easy to integrate and involves little or no CPU utilisation. Intel is the only manufacturer that currently integrates Ethernet support in all its motherboards.

✗ **Audio Modem Riser (AMR):** Integrates a low cost software modem onto the motherboard but the modem riser cards are hard to find and can cost as much as an internal PCI modem.

✗ **Integrated RAID support (Redundant Array of Inexpensive Disks):** While this feature promises to double the IDE transfer rates, the catch is that it functions only if you have two drives with identical characteristics and preferably from the same manufacturer. This limitation reduces its usefulness considerably.

✗ **Dual channel DDR support:** This technology can theoretically double the memory bandwidth, enabling AMD processors to outperform the Pentium 4/RDRAM combination. Unfortunately, it doesn't seem to work too well and the Athlon XP doesn't show any significant gains in real-world performance.

unpopulated RDRAM slots.

Our Verdict: The board is a great buy for performance enthusiasts and avid gamers.

HIS P4M266 (M50-12)

This board is an inexpensive Micro-ATX motherboard that uses the VIA P4M266 SDRAM chipset. Its performance was pretty decent with one of the best scores in the SDRAM category. In the SiSoft Sandra Drive Index, it notched an impressive score of 27395 points, and scored 5173 in the PCMark 2002 CPU Index.